ALEJANDRO W. SERE

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Applied mathematician completing an MSc in Systems and Control at TU Eindhoven (GPA: 9.1/10; expected cum laude), with experience in machine learning, optimization, and control. My work includes Recurrent Equilibrium Networks and using gradient-based learning to tune controllers; I have also trained neural networks for EEG-based emotion classification at Naxon Labs. I'm particularly interested in characterizing the limits of machine-learning models through theoretical analysis and empirical evaluation, including estimating their achievable performance in continual-learning settings.

EDUCATION

[2024-2026] Master of Science in Systems and Control at Eindhoven University of Technology, Netherlands. Degree conferral expected October 2026.
Expected to graduate **cum laude**, **GPA: 9.1/10**.

Specialization in **robust/optimal nonlinear control**, as well as in **networked systems**.

Tuned recurrent equilibrium network controllers using gradient descent during internship.

Graduation research thesis: **9/10**

- Topic: Reference tracking for nonlinear systems with performance and robustness guarantees.
- **Proposed different approaches** for system analysis and systematized control synthesis; including **gradient based approaches** for structured controller tuning.
- **Critically assessed** the viability of the proposed approaches via simulation studies and benchmarks.
- **Developed cutting-edge computational tool** for Scaled Relative Graphs in **Python**.
- Obtained knowledge on the **latest trends in robust nonlinear control**. **Research findings are being prepared for two publications**.

[2022-2024] Bachelor of Science in Applied Mathematics at Eindhoven University of Technology, Netherlands. **Cum laude**, **GPA: 9.1/10**.

Modeling projects include: modeling tsunamis using shallow water PDEs, biometric identification based on key-logger data, and finding the optimal way to check containers at a port for contraband.

Bachelor final project was about applying **model order reduction to a finite-element scheme simulation of Burgers' PDE**.

Specialization in **data mining and machine learning**, and **automotive sensing**.

Trained ML models to solve real problems with real datasets, including medical data and crime statistics.

Relevant courses: Probability, and Linear Algebra..

[2021, Jan-Feb] Internship at Naxon Labs.
Designed and trained neural networks to classify a subject's emotional state based on electroencephalography signals.

WORK EXPERIENCE

[2026, May-Jun] Teaching assistant for Master's courses at TU/e for Analysis and design of networked dynamical systems.
Supported students with exercises during practical lectures.

[2025, Nov-Dec] Teaching assistant for Master's courses at TU/e for Control principles for engineered systems.
Prepared practical sessions, handouts, lab setups, and the midterm. In charge of some practical sessions.

VOLUNTEER WORK

[2022-Present] Member of the Uruguayan Math Olympiad Problem Bank. This includes:

- **Creating original math olympiad problems**. Assessing the difficulty of olympiad-level problems. **Ensuring that proposed problems have valid solutions**.
- **Assembling math olympiad tests** for the Uruguayan National Olympiad, as well as team-selection tests for international competitions.
- **Correcting and grading said team-selection tests** to decide which students qualify for the competitions.
- **Co-author of two books from the Uruguayan Math Olympiad (fonmat)**:
  - **Libro Olimpiada Nacional 2018 y 2019**. Consisting of problems and solutions.
  - **Libro Campamento Matemático Uruguay 2026**. Discusses problem solving approaches and techniques.

[2020] Co-creator of UyMath!. A YouTube channel for math olympiad education in Spanish.

- Successfully created an online library so prospective olympiad students could study without needing a teacher.
- Took high level decisions, including which videos should cover which topics
- Edited half of the videos. Wrote, recorded and edited a third of the content.

SCIENCE COMPETITIONS AND AWARDS

[2018-2020] International Math Olympiad: Three participations, **2 honorable mentions**.

[2018-2019] Iberoamerican Math Olympiad: **Bronze and silver medals**.

[2018-2019] Uruguayan Math Olympiad: Gold Medal/**First place** (2 years in a row).

[2019] Uruguayan Physics Olympiad: **First place**. Iberoamerican Physics Olympiad **bronze medal**.

[2013-2024] More than 15 additional scientific olympiads, including 2 university competitions.

TECHNICAL SKILLS:

[2021-present] MATLAB (including system identification and LPVcore toolboxes), as well as **Simulink**

[2018-present] Python, including NumPy, SciPy, Jax, pandas, TensorFlow, Torch, CVXPY

[2021-present] Additionally: **Git**, Java, SQL. Mercor's ***introduction to agents*** course (2026).

HOBBIES AND INTERESTS

Boxing: Enjoy sparring and adapting tactics in real time.

Cooking: Developing my skills and exploring how food brings people together.

Video games: Enjoying challenging roguelikes and refining strategies through trial and error.

LANGUAGES

Spanish: Native, English: Fluent.